

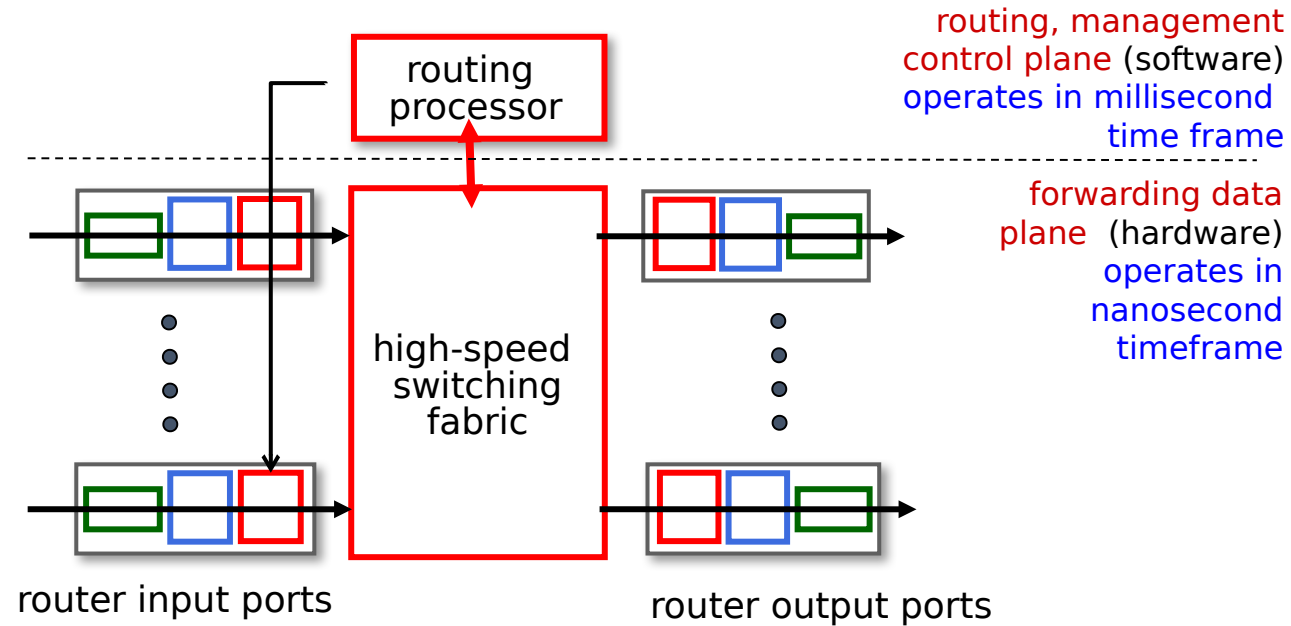
Computer Networks

Router Architecture and Scheduling

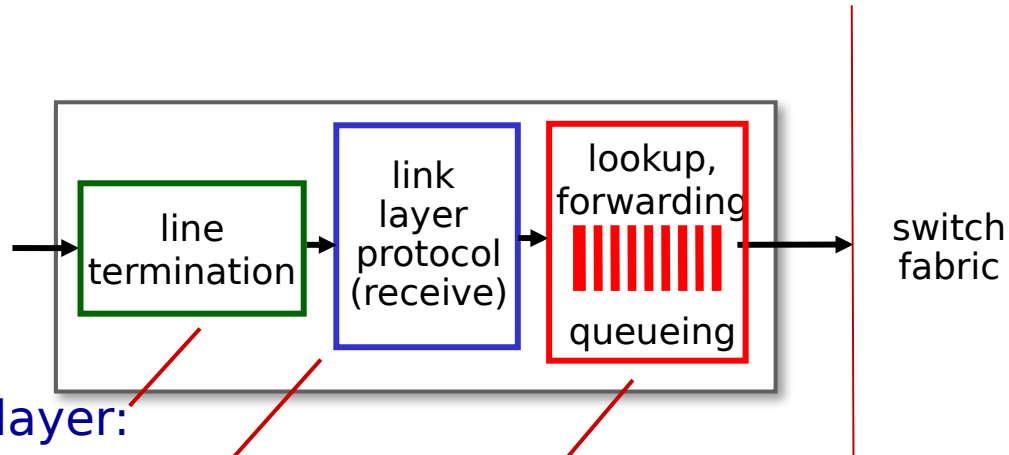
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Router Architecture

Router architecture overview



Input port functions

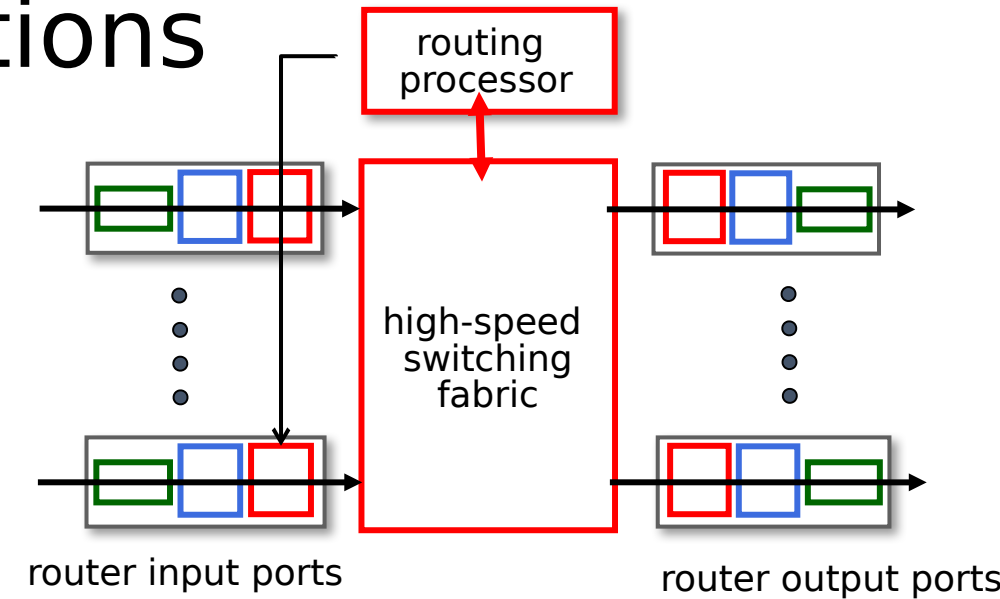


Physical layer:
Bit-level reception

Link layer:
e.g., Ethernet

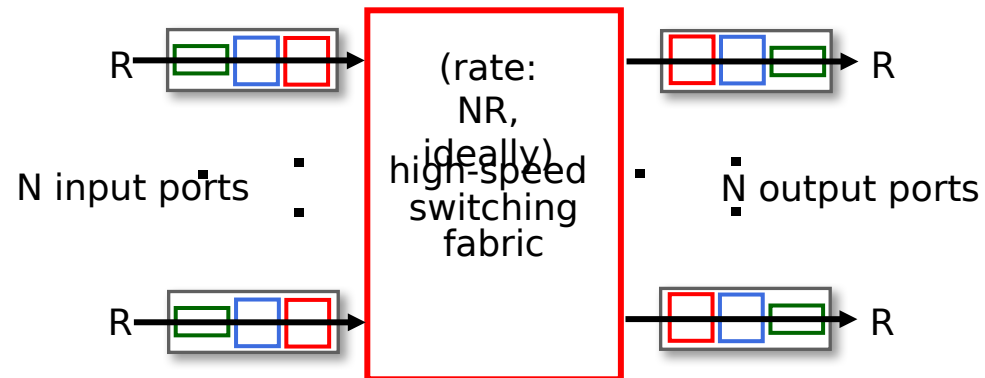
Decentralized switching:

- Using header field values, lookup output port using forwarding table in input port memory (“match plus action”)
- Goal: complete input port processing at ‘line speed’
- **Input port queueing**: if datagrams arrive faster than forwarding rate into switch fabric



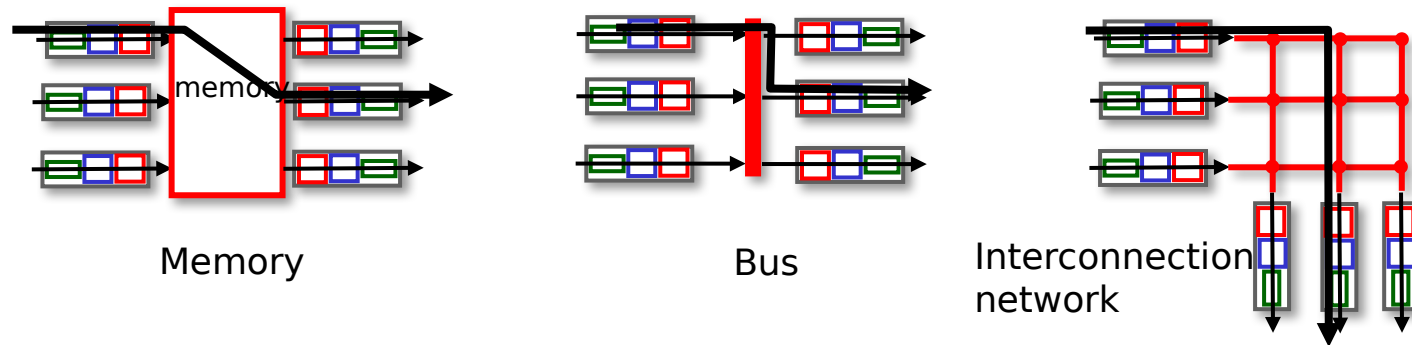
Switching fabrics

- Transfer packet from input link to appropriate output link
- **Switching rate:** rate at which packets can be transfer from inputs to outputs
 - Often measured as multiple of input/output line rate
 - N inputs: switching rate N times line rate desirable



Switching fabrics

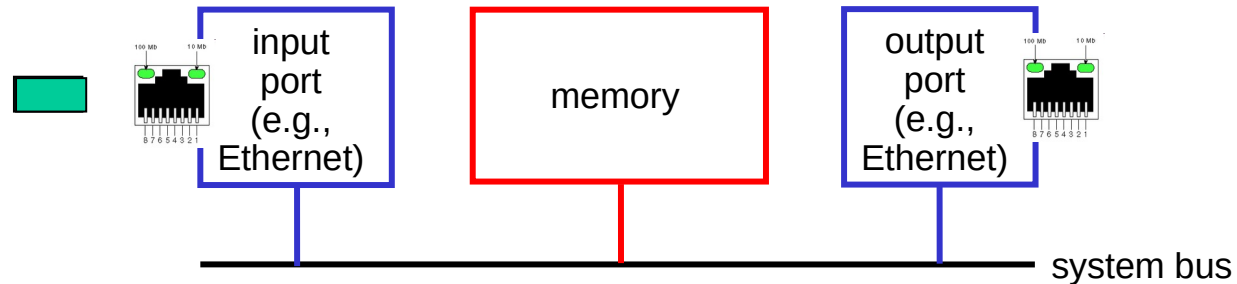
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- **Switching rate:** rate at which packets can be transfer from inputs to outputs
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- Three major types of switching fabrics:



Switching via memory

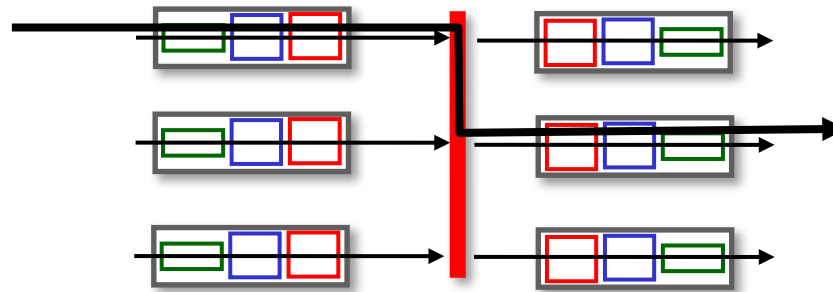
First generation routers:

- Traditional computers with switching under direct control of CPU
- Packet copied to system's memory
- Speed limited by memory bandwidth (2 bus crossings per datagram)



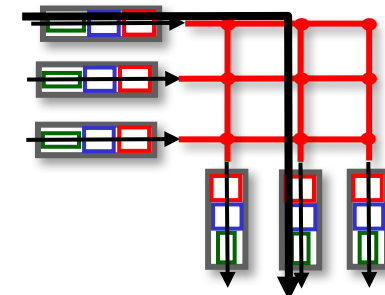
Switching via a bus

- Datagram from input port memory to output port memory via a shared bus
- **Bus contention:** switching speed limited by bus bandwidth
- 32 Gbps bus, Cisco 5600: sufficient speed for access routers

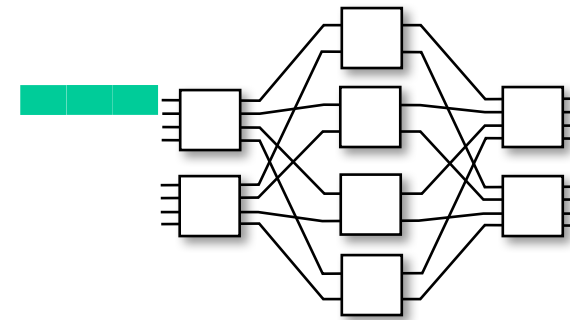


Switching via interconnection network

- Crossbar, Clos networks, other interconnection nets initially developed to connect processors in multiprocessor computer architecture
- **Multistage switch:** $n \times n$ switch from multiple stages of smaller switches
- **Exploiting parallelism:**
 - Fragment datagram into fixed length cells on entry
 - Switch cells through the fabric, reassemble datagram at exit



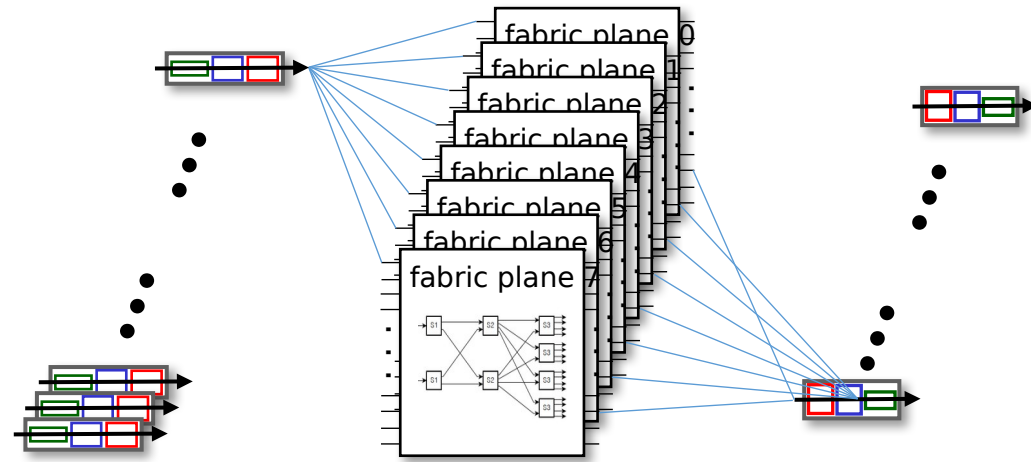
3x3 crossbar



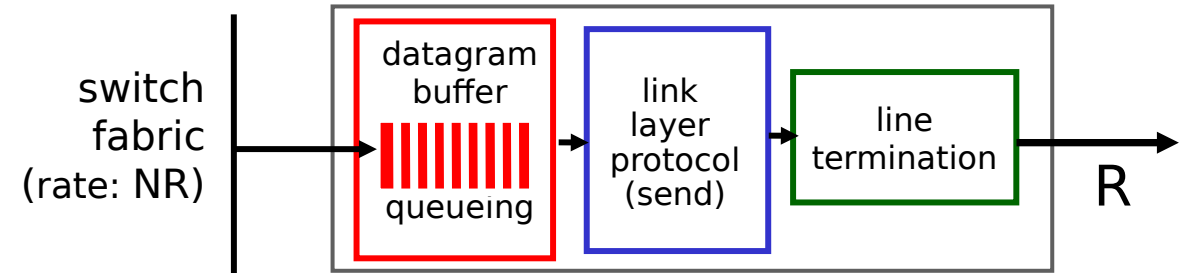
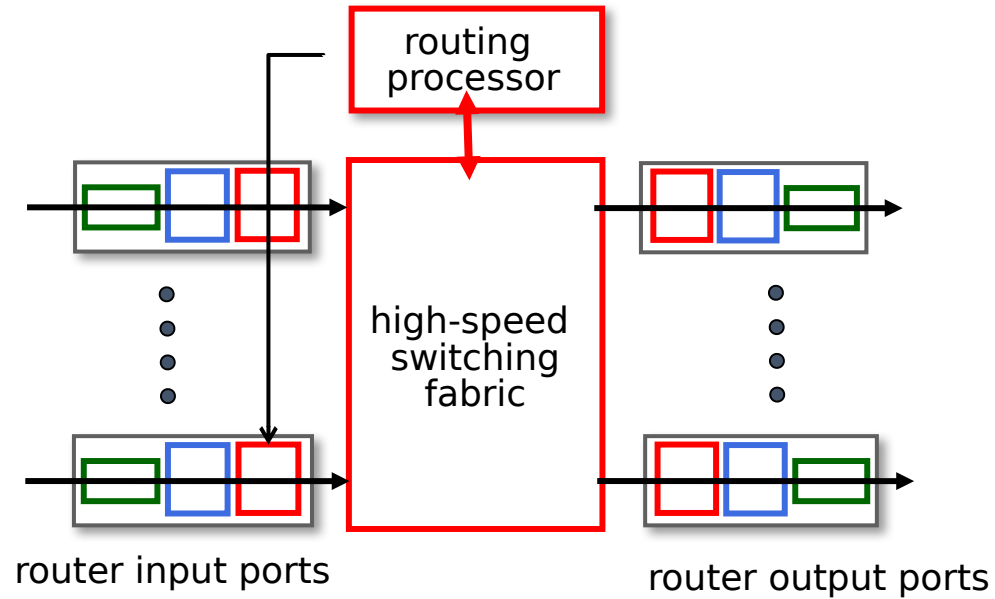
8x8 multistage switch
built from smaller-sized switches

Switching via interconnection network

- Scaling, using multiple switching “planes” in parallel:
 - Speedup, scaleup via parallelism
- Cisco CRS router:
 - Basic unit: 8 switching planes
 - Each plane: 3-stage interconnection network
 - up to 100's Tbps switching capacity
- Cisco CRS router: <https://nexstor.com/wp-content/uploads/2018/05/cisco-crs-1-multishelf-system-datasheet.pdf>



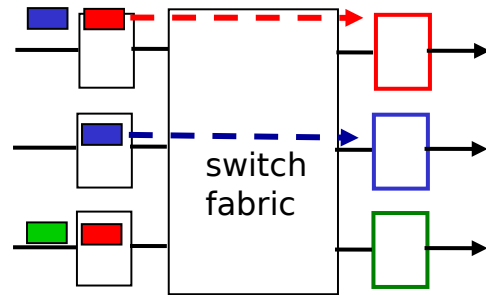
Output port functions



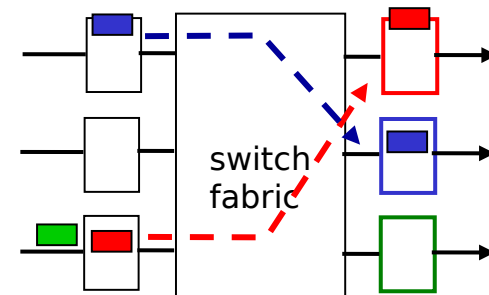
Queuing, Buffer management and Scheduling

Input port queuing

- If switch fabric slower than input ports combined → queueing may occur at input queues
 - Queueing delay and loss due to input buffer overflow!
- **Head-of-the-Line (HOL) blocking:** Queued datagram at front of queue prevents others in queue from moving forward

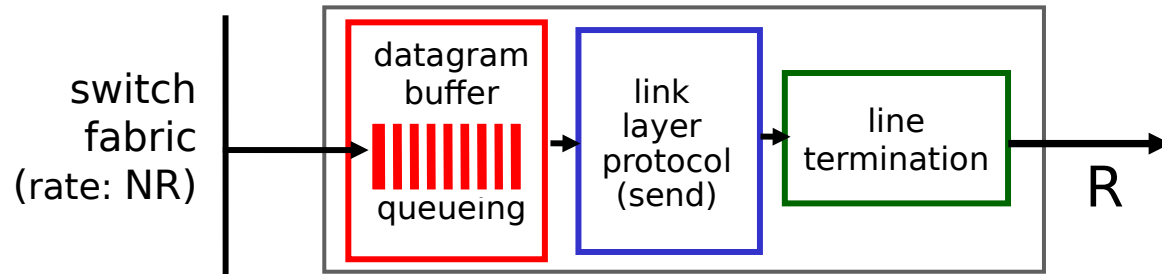


Output port contention: only one red datagram can be transferred. lower red packet is **blocked**



One packet time later:
green packet experiences HOL blocking

Output port queuing



- **Buffering** required when datagrams arrive from fabric faster than link transmission rate
- **Drop policy:** which datagrams to drop if no free buffers?
- **Scheduling discipline** chooses among queued datagrams for transmission



Datagrams can be lost due to congestion, lack of buffers



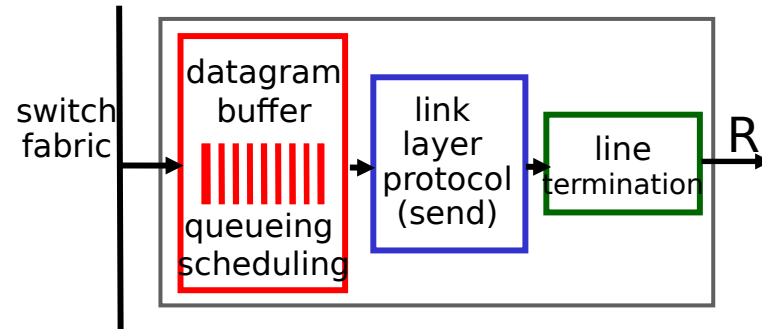
Priority scheduling – who gets best performance

How much buffering?

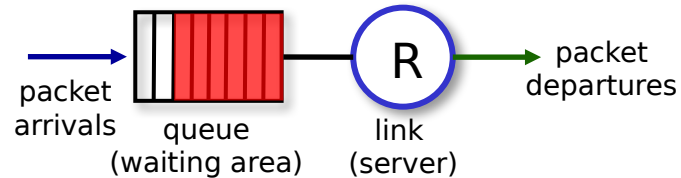
- Too much buffering will reduce packet loss, but can increase delays
 - Long RTTs: poor performance for real-time apps, sluggish TCP response
- RFC 3439 rule of thumb: average buffering equal to “typical” RTT times link capacity C
 - e.g., RTT = 250 msec, C = 10 Gbps link → 2.5 Gbit buffer
 - **Delay-bandwidth product**
- More recent recommendation: with N flows, buffering equal to

$$\frac{RTT \cdot C}{\sqrt{N}}$$

Buffer Management



Abstraction: queue



Buffer management:

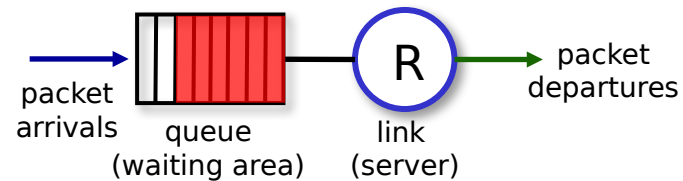
- **Drop:** which packet to add, drop when buffers are full
 - **Tail drop:** drop arriving packet
 - **Priority:** drop/remove on priority basis
- **Marking:** which packets to mark to signal congestion (i.e. ECN)

Packet Scheduling: FCFS

Packet scheduling:
deciding which packet
to send next on link

- First come, first served
- Priority based
- Round robin
- Weighted fair queueing

Abstraction: queue



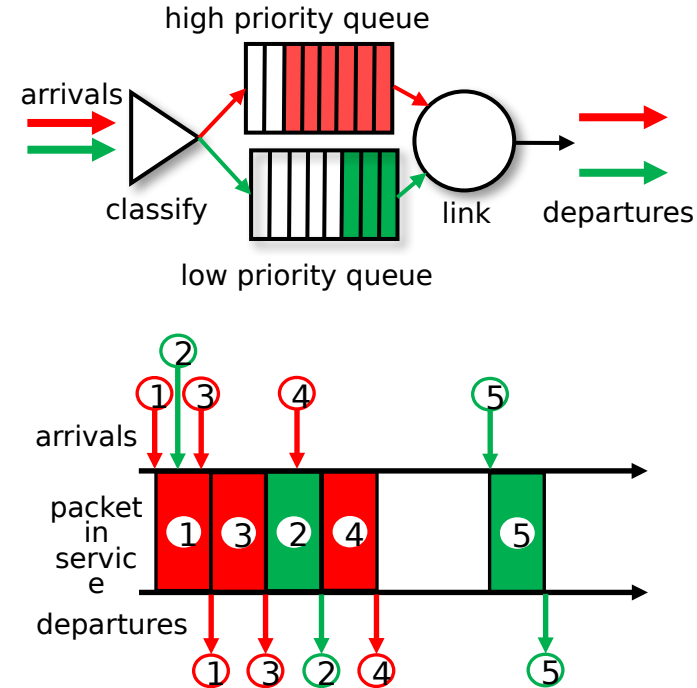
FCFS: packets
transmitted in order
of arrival to output
port

- also known as: First-in-first-out (FIFO)

Scheduling policies: Priority Based

Priority based scheduling:

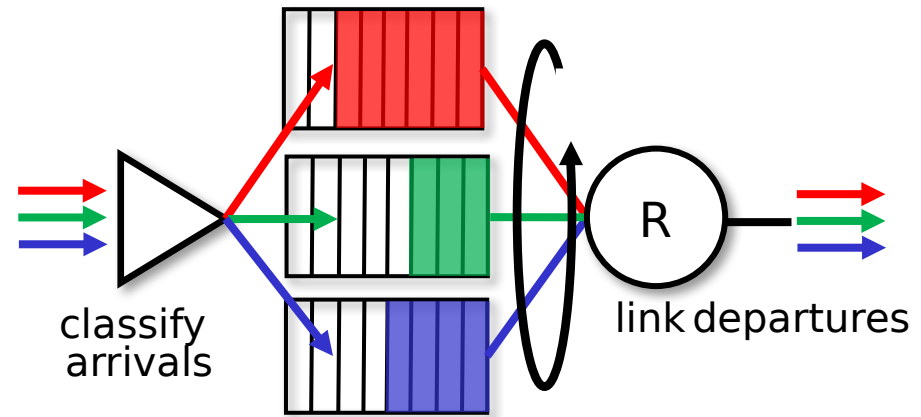
- Arriving traffic classified, queued by class
 - Any header fields can be used for classification
- Send packet from highest priority queue that has buffered packets
 - FCFS within priority class



Scheduling policies: round robin

Round Robin (RR) scheduling:

- Arriving traffic classified, queued by class
 - Any header fields can be used for classification
- Server cyclically, repeatedly scans class queues, sending one complete packet from each class (if available) in turn



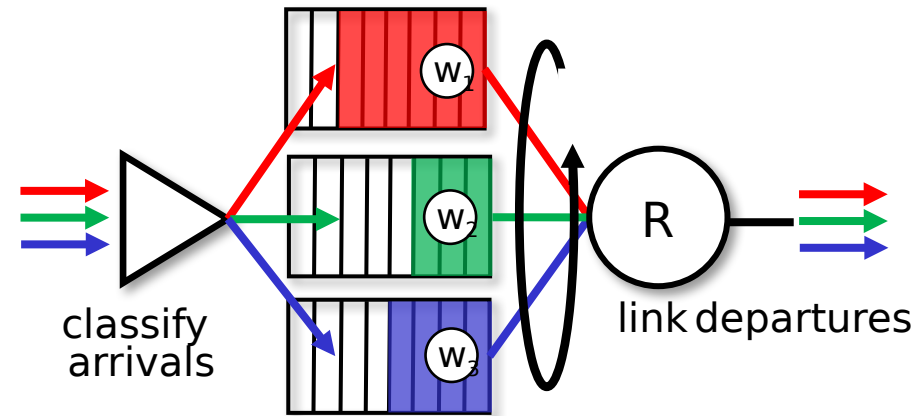
Scheduling policies: Weighted Fair Queueing

Weighted Fair Queueing (WFQ):

- Generalized Round Robin
- Each class i , has weight, w_i and gets weighted amount of service in each cycle:

$$\frac{w_i}{\sum_j w_j}$$

- Minimum bandwidth guarantee (per-traffic-class)



Summary

□ Router architecture, queuing and packet scheduling:

- Router architecture
 - Input ports
 - High speed fabric
 - Out ports
 - Packet scheduling
 - FCFS
 - Priority based
 - Round robin
 - Weighted fair queuing
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